

Hrsh Venket

<https://github.com/Hrsh-Venket>

<https://hrsh-venket.github.io>

hrsh.venket@gmail.com

AI Engineer with experience in computer vision, machine learning, and large-scale data analysis, specializing in time series, remote sensing, and Computer Vision. Skilled in building scalable ML pipelines, training and evaluating deep learning models (PyTorch, TensorFlow), and optimizing algorithms for real-world deployment. Published and presented work at NASA SARI Conference; paper under journal review. Seeking to apply expertise in AI, computer vision, and distributed systems to develop cutting-edge models and deploy them at scale.

Work Experience

Research Assistant

Aug 2023 - Jul 2025

Mphasis AI and Applied Tech Lab at Ashoka

First author for paper under journal submission. Trained models for satellite time series data, improving Indian forest fire classification accuracy by 9% over SoTA. Conducted large-scale data preprocessing, feature engineering, and validation for remote sensing datasets.

Presented working paper at NASA SARI Conference 2024 to top delegates: Dr. David Skole (Nobel Laureate) and Dr. Krishna Vadrevu (NASA).

Data Science Intern

Oct 2023 - Dec 2024

Tiger Analytics Pvt. Limited

Developed and optimized Bayesian regression, additive, and multiplicative ML models for market mix modeling.

Applied feature engineering, validation, and numerical optimization to analyze sales drivers and improve forecasting accuracy.

Teaching Assistant (2 semesters)

Aug 2023 - Dec 2024

Ashoka University

CS-3410: *Introduction to Machine Learning* in 2023, CS-4912: *Computer Vision* in 2024. 120+ students and 50+ Office hours.

Projects

Advanced TFT Analytics Tool Aug 2024 - Present (Scan the QR code to check it out)

Designed and deployed a web app with autoscaling on Google Cloud Platform, backed by BigQuery for large-scale data processing. Currently (LLMs) to generate human readable and actionable insights from ingested data

Built an automated data ingestion pipeline from scratch to ensure seamless, reliable integration of incoming data.

Capstone Project: Predicting Forest Fire spread in Central India

Aug 2024 - Dec 2024

Replicated DiffusionSat (Diffusion model specific to Satellite Data) to predict forest fire spread in central India

Applied Monte Carlo Methods and to predict forest fire spread.

3D Reconstruction of C Elegans

Jan 2024 - May 2024

Built a Computer Vision pipeline that involved image processing, 3D reconstruction and visualisation of *C. Elegans* from SDCM images.

Load Forecasting Bounty - Machine Learning, Electricity Load Forecasting

Jan 2024 - May 2024

Applied gradient boosted trees and time series encoding techniques for large-scale electricity load forecasting.

Achieved 24th/195 teams (Team: Hrsh Venket on [public leaderboard](#))



Technical Skills

Programming & Domains: Python, C/C++, PyTorch, TensorFlow, OpenCV, Computer Vision, Machine Learning, Deep Learning, Remote Sensing,

Time Series Analysis, Numerical Optimization, Image Processing, Algorithms, Data Structures, Statistics, High-Performance Computing

Platforms: Google Cloud Platform, BigQuery, Unix/Linux, Riot Developer API, Docker, Google Earth Engine

Education

The Hong Kong University of Science and Technology

Sep 2025 - Present

Master of Science in Artificial Intelligence

Ashoka University

Aug 2021 - Jun 2025

Postgraduate Diploma in Advanced Studies and Research, Advanced Major in Computer Science

Key Classes: Intro to Machine Learning, Computer Vision, Numerical Algorithms and Optimisation, Computer Graphics, Statistical Inference

APL Global School

Jan 2017 - Jan 2021

A levels: Received As in Economics, Mathematics, Physics, Classical Studies

Leadership

Captain of Men's Volleyball Team Ashoka University

Aug 2024 - May 2025

Rebuilt squad after half of the team graduated, Captained 18 over 9 tournaments.

President of Eeshto (the Board Gaming Society) Ashoka University

Jan 2023 - Jan 2024

Managed a team of 50 across 6 departments, doubled membership, organised 20+ Campus events, won award for contributions to campus life

School Pupil Leader, Secretary General APL Global School

Jul 2020 - Jun 2021